Ideation Phase

Empathize & Discover

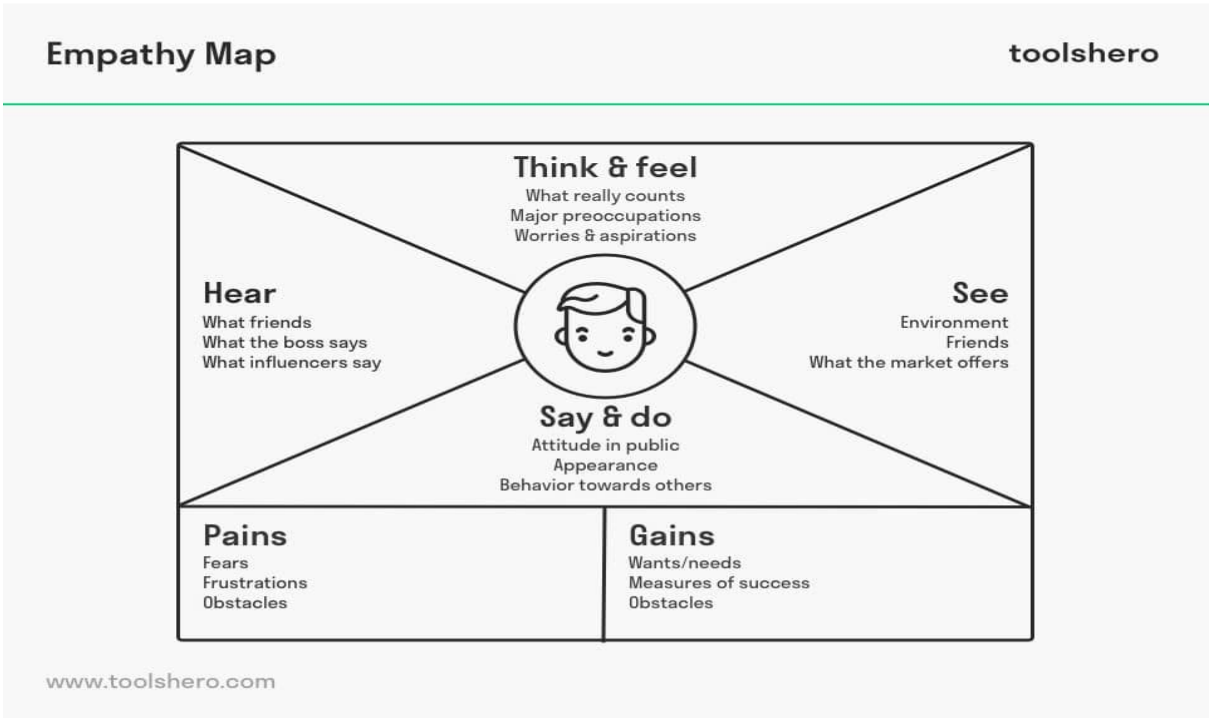
Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Empathy Map Canvas

An **Empathy Map Canvas** is a visual tool used to understand the needs, thoughts, feelings, behaviors, challenges, and expectations of the end users. It helps the project team view the problem from the user's perspective and design a solution that addresses real-world requirements.

For the **Electricity Consumption Analysis** project, the empathy map focuses on understanding the needs of energy analysts, electricity department officials, and decision-makers who analyze electricity consumption data. It identifies their goals, pain points, and expectations while monitoring electricity usage across different states and regions.

By creating the empathy map, the team gained a better understanding of user requirements such as interactive dashboards, state-wise comparisons, monthly trend analysis, and effective visualization of electricity consumption data. This understanding helped in designing a Tableau dashboard that provides meaningful insights and supports better energy planning and decision-making.



Our Empathy Map Canvas: Electricity Consumption Analysis

Primary user persona:

An energy analyst or government electricity department officer who regularly monitors electricity consumption across different states and regions to support efficient energy planning, demand forecasting, and resource allocation.

THINKS

- Wonders which state has the highest electricity consumption.
- Wants accurate monthly electricity usage reports.
- Thinks about improving energy distribution.
- Considers how electricity demand changes across seasons.
- Needs reliable data for planning future electricity requirements.

SEES

- Large electricity consumption datasets.
- Different electricity demand across states.
- Seasonal fluctuations in electricity usage.
- Regional differences in power consumption.
- Growing electricity demand due to urbanization.

HEARS

- Government energy policies.
- Electricity board reports.
- Feedback from consumers.
- Suggestions from energy experts.
- News about electricity shortages and demand.

SAYS

- "We need accurate electricity usage reports."
- "The dashboard should clearly compare all states."
- "Monthly trends help us plan better."
- "Interactive visualization makes analysis easier."

DOES

- Collects electricity consumption data.
- Compares state-wise electricity usage.
- Monitors monthly trends.
- Generates reports for decision-makers.
- Uses Tableau dashboards to analyze electricity demand.

PAINS

- Manual analysis is time-consuming.
- Difficult to compare multiple states.
- Large datasets are hard to interpret.
- Traditional reports lack visualization.
- Delayed business decisions due to poor reporting.

GAINS • Interactive Tableau dashboards. • Faster electricity consumption analysis. • Better comparison between states and regions. • Improved resource planning. • Data-driven decision making. • Easy identification of high-demand areas.

User Needs

- Accurate electricity consumption analysis.
- Interactive dashboards with filters.
- State-wise and region-wise comparison.
- Monthly electricity usage trends.
- Geographic visualization using maps.
- Quick report generation.
- Reliable business insights for planning.

Opportunities

- Improve electricity demand forecasting.
- Support government energy planning.
- Optimize electricity resource allocation.
- Identify high electricity-consuming regions.
- Enable data-driven policy decisions.
- Expand the project using machine learning for future demand prediction.